Section 1 Pain

# 14 Pain: Pathophysiology and Management

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The province of medicine is to preserve and restore health and to relieve suffering. Understanding pain is essential to both goals. Because pain is universally understood as a signal of disease, it is the most common symptom that brings a patient to a physician's attention. The function of the pain sensory system is to protect the body and maintain homeostasis. It does this by detecting, localizing, and identifying potential or actual tissue-damaging processes. Because different diseases produce characteristic patterns of tissue damage, the quality, time course, and location of a patient's pain lend important diagnostic clues. It is the physician's responsibility to assess each patient promptly for any remediable cause underlying the pain and to provide rapid and effective pain relief whenever possible.

## THE PAIN SENSORY SYSTEM

Pain is an unpleasant sensation localized to a part of the body. It is often described in terms of a penetrating or tissue-destructive process (e.g., stabbing, burning, twisting, tearing, squeezing) and/or of a bodily or emotional reaction (e.g., terrifying, nauseating, sickening). Furthermore, any pain of moderate or higher intensity is accompanied by anxiety and the urge to escape or terminate the feeling. These properties illustrate the duality of pain: it is both sensation and emotion. When it is acute, pain is characteristically associated with behavioral arousal and a stress response consisting of increased blood pressure, heart rate, pupil diameter, and plasma cortisol levels. In addition, local muscle contraction (e.g., limb flexion, abdominal wall rigidity) is often present.

### PERIPHERAL MECHANISMS

**The Primary Afferent Nociceptor** A peripheral nerve consists of the axons of three different types of neurons: primary sensory afferents, motor neurons, and sympathetic postganglionic neurons (Fig. 14-1). The cell bodies of primary sensory afferents are in the dorsal root ganglia within the vertebral foramina. The primary afferent axon has two branches: one projects centrally into the spinal cord and the other projects peripherally to innervate tissues. Primary afferents are classified by their diameter, degree of myelination, and conduction velocity. The largest diameter afferent fibers, A-beta (Aβ), respond maximally to light touch and/or moving stimuli; they are present primarily in nerves that innervate the skin. In normal individuals, the activity of these fibers does not produce pain. There are two other classes of primary afferent nerve fibers: the small diameter myelinated A-delta (Aδ) and the unmyelinated (C) axons (Fig. 14-1). These fibers are present in nerves to the skin and to deep somatic and visceral structures. Some tissues, such as the cornea, are innervated only by Aδ and C fiber afferents. Most Aδ and C fiber afferents respond maximally to intense (painful) stimuli and produce the subjective experience of pain when they are activated; this defines them as *primary afferent nociceptors* (*pain receptors*). The ability to detect painful stimuli is completely abolished when conduction in Aδ and C fiber axons is blocked.

Individual primary afferent nociceptors can respond to several different types of noxious stimuli. For example, most nociceptors respond to heat; intense cold; intense mechanical distortion, such as a pinch; changes in pH, particularly an acidic environment; and application of chemical irritants including adenosine triphosphate (ATP), serotonin, bradykinin (BK), and histamine. The transient receptor potential cation channel subfamily V member 1 (TRPV1), also known as the vanilloid receptor, mediates perception of some noxious stimuli, especially heat sensations, by nociceptive neurons; it is activated by heat, acidic pH, endogenous mediators, and capsaicin, a component of hot chili peppers.

**Sensitization** When intense, repeated, or prolonged stimuli are applied to damaged or inflamed tissues, the threshold for activating primary afferent nociceptors is lowered, and the frequency of firing is higher for all stimulus intensities. Inflammatory mediators such as BK, nerve-growth factor, some prostaglandins (PGs), and leukotrienes contribute to this process, which is called *sensitization*. Sensitization occurs at the level of the peripheral nerve terminal (*peripheral sensitization*) as well as at the level of the dorsal horn of the spinal cord (*central sensitization*). Peripheral sensitization occurs in damaged or inflamed tissues, when inflammatory mediators activate intracellular signal transduction in nociceptors, prompting an increase in the production, transport, and membrane insertion of chemically gated and voltage-gated ion channels. These changes increase the excitability of nociceptor terminals and lower their threshold for activation by mechanical, thermal, and chemical stimuli. Central sensitization occurs when activity, generated by nociceptors during inflammation, enhances the excitability of nerve cells in the dorsal horn of the spinal cord. Following injury and resultant sensitization, normally innocuous stimuli can produce pain (termed *allodynia*). Sensitization is a clinically important process that contributes to tenderness, soreness, and *hyperalgesia* (increased pain intensity in response to the same noxious stimulus; e.g., pinprick causes severe pain). A striking example of sensitization is sunburned skin, in which severe pain can be produced by a gentle slap or a warm shower.

Sensitization is of particular importance for pain and tenderness in deep tissues. Viscera are normally relatively insensitive to noxious mechanical and thermal stimuli, although hollow viscera do generate

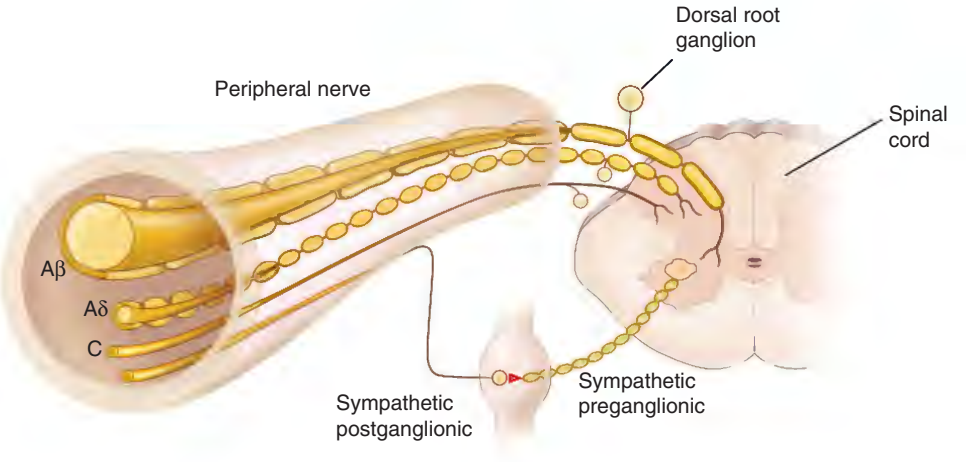


**FIGURE 14-1 Components of a typical cutaneous nerve.** There are two distinct functional categories of axons: primary afferents with cell bodies in the dorsal root ganglion and sympathetic postganglionic fibers with cell bodies in the sympathetic ganglion. Primary afferents include those with large-diameter myelinated (Aβ), small-diameter myelinated (Aδ), and unmyelinated (C) axons. All sympathetic postganglionic fibers are unmyelinated.

significant discomfort when distended. In contrast, when affected by a disease process with an inflammatory component, deep structures such as joints or hollow viscera characteristically become exquisitely sensitive to mechanical stimulation.

A large proportion of Aδ and C fiber afferents innervating viscera are completely insensitive in normal noninjured, noninflamed tissue. That is, they cannot be activated by known mechanical or thermal stimuli and are not spontaneously active. However, in the presence of inflammatory mediators, these afferents become sensitive to mechanical stimuli. Such afferents have been termed *silent nociceptors*, and their characteristic properties may explain how, under pathologic conditions, the relatively insensitive deep structures can become the source of severe and debilitating pain and tenderness. Low pH, PGs, leukotrienes, and other inflammatory mediators such as BK play a significant role in sensitization.

**Nociceptor-Induced Inflammation** Primary afferent nociceptors are not simply passive messengers of threats to tissue injury but also play an active role in tissue protection through a neuroeffector function. Most nociceptors contain polypeptide mediators, including substance P, calcitonin gene related peptide (CGRP), and cholecystokinin, that are released from their peripheral terminals when they are activated **(Fig. 14-2)**. Substance P is an 11-amino-acid peptide that is released in peripheral tissues from primary afferent nociceptors and has multiple biologic activities. It is a potent vasodilator, causes mast cell degranulation, is a chemoattractant for leukocytes, and increases the production and release of inflammatory mediators. Interestingly, depletion of substance P from joints reduces the severity of experimental arthritis.

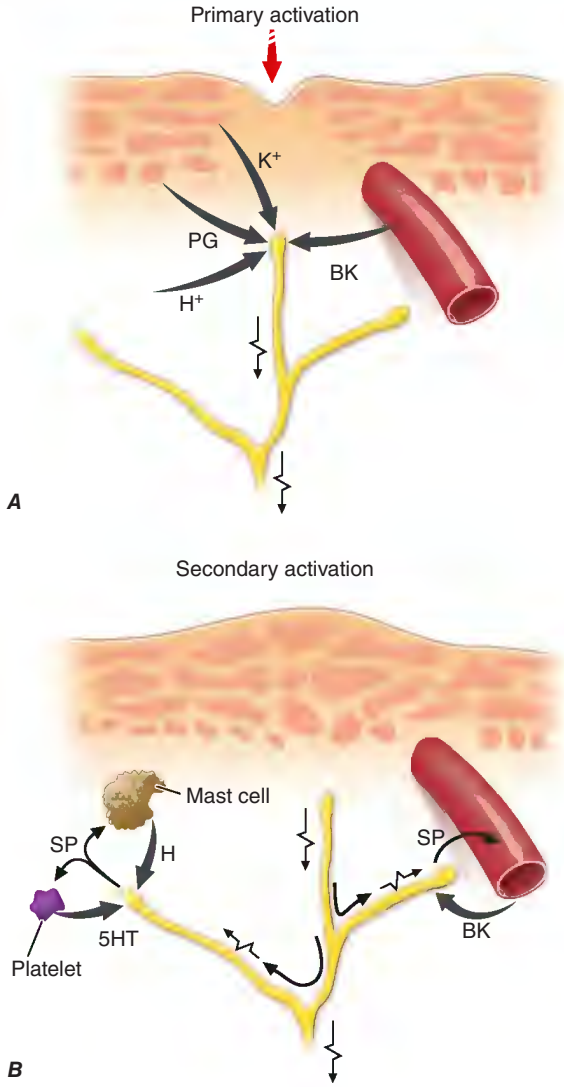


**FIGURE 14-2 Events leading to activation, sensitization, and spread of sensitization of primary afferent nociceptor terminals. *A*.** Direct activation by intense pressure and consequent cell damage. Cell damage induces lower pH ($H^+$) and leads to release of potassium ($K^+$) and to synthesis of prostaglandins (PGs) and bradykinin (BK). PGs increase the sensitivity of the terminal to BK and other pain-producing substances. ***B*.** Secondary activation. Impulses generated in the stimulated terminal propagate not only to the spinal cord but also into other terminal branches where they induce the release of peptides, including substance P (SP). Substance P causes vasodilation and neurogenic edema with further accumulation of BK. Substance P also causes the release of histamine (H) from mast cells and serotonin (5HT) from platelets.

## CENTRAL MECHANISMS

**The Spinal Cord and Referred Pain** The axons of primary afferent nociceptors enter the spinal cord via the dorsal root. They terminate in the dorsal horn of the spinal gray matter **(Fig. 14-3)**. The terminals of primary afferent axons contact spinal neurons that transmit the pain signal to brain sites involved in pain perception. When primary afferents are activated by noxious stimuli, they release neurotransmitters from their terminals that excite the spinal cord neurons. The major neurotransmitter released is glutamate, which rapidly excites the second-order dorsal horn neurons. Primary afferent nociceptor terminals also release substance P and CGRP, which produce a slower and longer-lasting excitation of the dorsal horn neurons. The axon of each primary afferent contacts many spinal neurons, and each spinal neuron receives convergent inputs from many primary afferents.

The convergence of sensory inputs to a single spinal pain-transmission neuron is of great importance because it underlies the phenomenon of referred pain. All spinal neurons that receive input from the viscera and deep musculoskeletal structures also receive input from the skin. The convergence patterns are determined by the spinal segment of the dorsal root ganglion that supplies the afferent innervation of a structure. For example, the afferents that supply the central diaphragm are derived from the third and fourth cervical dorsal root ganglia. Primary afferents with cell bodies in these same ganglia supply the skin of the shoulder and lower neck. Thus, sensory inputs from both the shoulder skin and the central diaphragm converge on pain-transmission neurons in the third and fourth cervical spinal segments. *Because of this convergence and the fact that the spinal neurons are most often activated by inputs from the skin, activity evoked in spinal neurons by input from deep structures is often mislocalized by the patient to a bodily location that roughly corresponds with the region of skin innervated by the same spinal segment*. Thus, inflammation near the central diaphragm is often reported as shoulder discomfort. This spatial displacement of pain sensation from the site of the injury that produces it is known as *referred pain*.

**Ascending Pathways for Pain** Most spinal dorsal horn neurons activated by primary afferent nociceptors send their axons to the contralateral thalamus. These axons form the contralateral spinothalamic tract, which lies in the anterolateral white matter of the spinal cord, the lateral edge of the medulla, and the lateral pons and midbrain. The spinothalamic pathway is crucial for pain sensation in humans. Interruption of this pathway produces permanent deficits in pain and temperature discrimination.

Spinothalamic tract axons ascend to several regions of the thalamus. There is significant divergence of the pain signal from these thalamic sites to several distinct areas of the cerebral cortex that subserve different aspects of the pain experience **(Fig. 14-4)**. One of the thalamic projections is to the somatosensory cortex. This projection mediates the sensory discriminative aspects of pain, i.e., its location, intensity, and quality. Other thalamic neurons project to cortical regions that are linked to emotional responses, such as the anterior cingulate and insular cortex. These pathways to the cortex subserve the affective or unpleasant emotional dimension of pain. This affective dimension of pain produces suffering and exerts potent control of behavior. Because of this dimension, fear is a constant companion of pain. Consequently, injury or surgical lesions to areas of the frontal cortex activated by painful stimuli can diminish the emotional

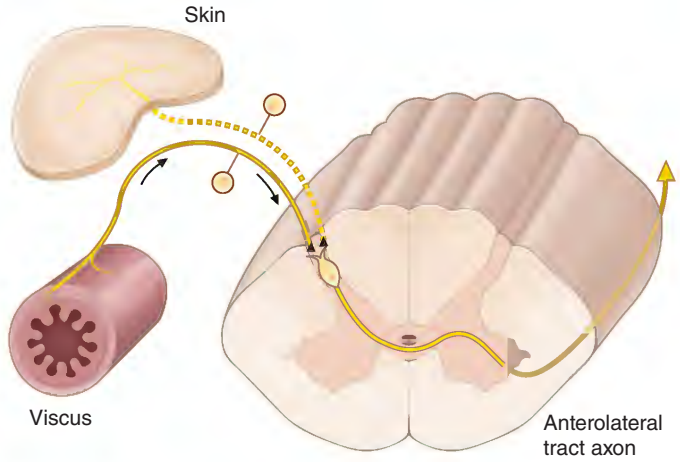


**FIGURE 14-3 The convergence-projection hypothesis of referred pain.** According to this hypothesis, visceral afferent nociceptors converge on the same pain-projection neurons as the afferents from the somatic structures in which the pain is perceived. The brain has no way of knowing the actual source of input and mistakenly "projects" the sensation to the somatic structure.

impact of pain while largely preserving the individual's ability to localize and recognize stimuli as painful.

## PAIN MODULATION

The pain produced by injuries of similar magnitude is remarkably variable in different situations and in different individuals. For example, athletes have been known to sustain serious fractures with only minor pain, and Beecher's classic World War II survey revealed that many soldiers in battle were unbothered by injuries that would have produced agonizing pain in civilian patients. Furthermore, even the suggestion that a treatment will relieve pain can have a significant analgesic effect (the *placebo effect*). On the other hand, many patients find even minor injuries such as venipuncture frightening and unbearable, and the expectation of pain can induce pain even without a noxious stimulus. The suggestion that pain will worsen following administration of an inert substance can increase its perceived intensity (the *nocebo effect*).

The powerful effect of expectation and other psychological variables on the perceived intensity of pain is explained by brain circuits that modulate the activity of the pain-transmission pathways. One of these circuits has links to the hypothalamus, midbrain, and medulla, and it selectively controls spinal dorsal horn pain-transmission neurons through a descending pathway (Fig. 14-4).

Human brain-imaging studies have implicated this pain-modulating circuit in the pain-relieving effect of attention, suggestion, and opioid analgesic medications (Fig. 14-5). Furthermore, each of the component structures of the pathway contains opioid receptors and is sensitive to the direct application of opioid drugs. In animals, lesions of this

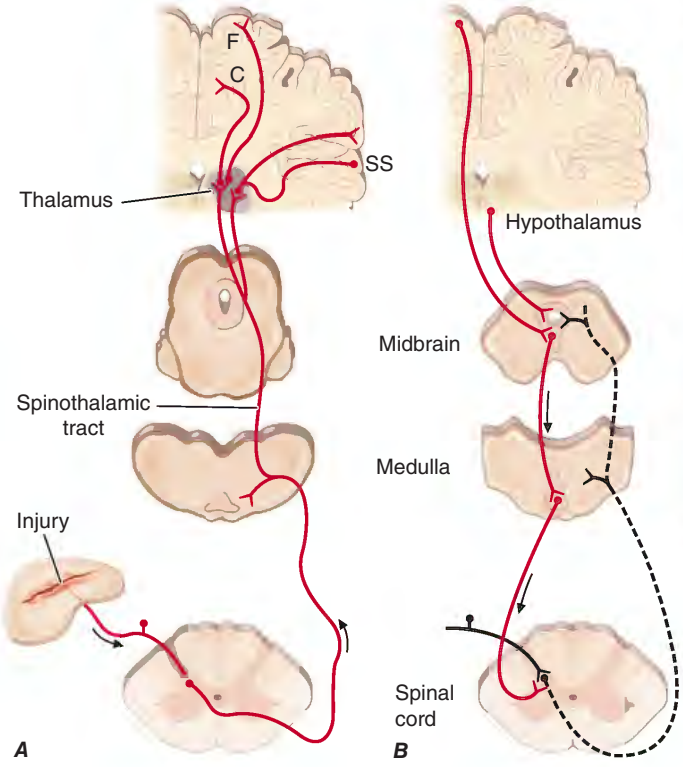


**FIGURE 14-4 Pain-transmission and modulatory pathways. *A*.** Transmission system for nociceptive messages. Noxious stimuli activate the sensitive peripheral ending of the primary afferent nociceptor by the process of transduction. The message is then transmitted over the peripheral nerve to the spinal cord, where it synapses with cells of origin of the major ascending pain pathway, the spinothalamic tract. The message is relayed in the thalamus to the anterior cingulate (C), frontal insular (F), and somatosensory cortex (SS). ***B***. Pain-modulation network. Inputs from frontal cortex and hypothalamus activate cells in the midbrain that control spinal pain-transmission cells via cells in the medulla.

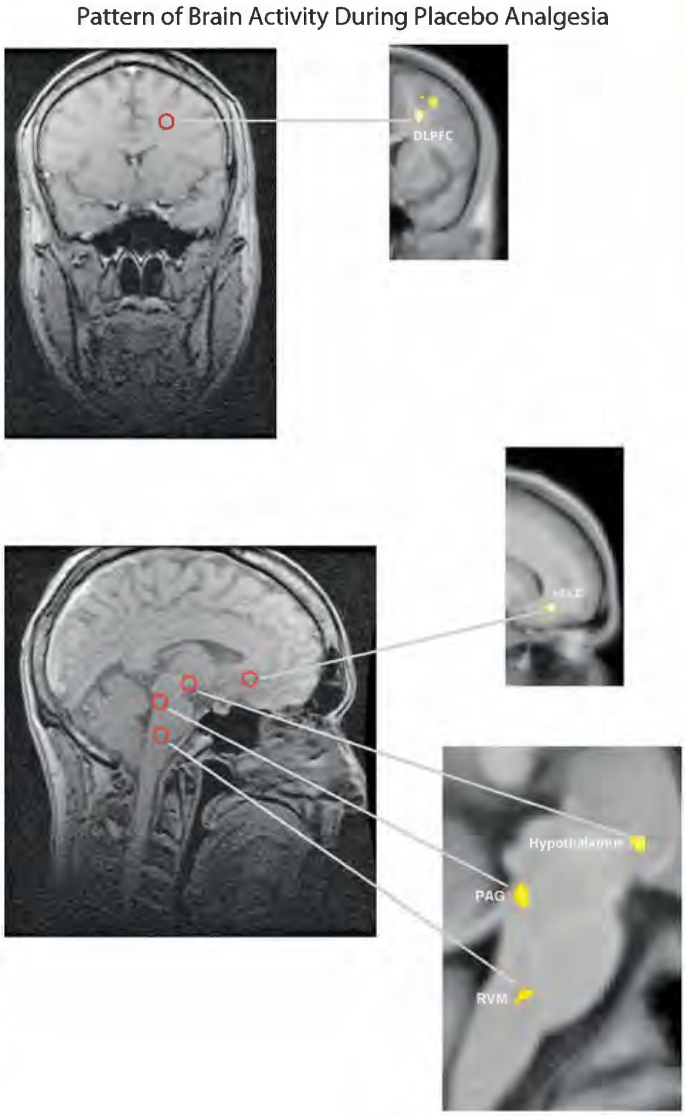


**FIGURE 14-5 Functional magnetic resonance imaging (fMRI) demonstrates placebo-enhanced brain activity in anatomic regions correlating with the opioidergic descending pain control system.** *Top panel:* Frontal fMRI image shows placebo-enhanced brain activity in the dorsal lateral prefrontal cortex (DLPFC). *Bottom panel:* Sagittal fMRI images show placebo-enhanced responses in the rostral anterior cingulate cortex (rACC), the rostral ventral medullae (RVM), the periaqueductal gray (PAG) area, and the hypothalamus. The placebo-enhanced activity in all areas was reduced by naloxone, demonstrating the link between the descending opioidergic system and the placebo analgesic response. *(Reproduced with permission from F Eippert et al: Activation of the opioidergic descending pain control system underlies placebo analgesia. Neuron 63:533, 2009.)*

descending modulatory system reduce the analgesic effect of systemically administered opioids such as morphine. Along with the opioid receptor, the component nuclei of this pain-modulating circuit contain endogenous opioid peptides such as the enkephalins and β-endorphin.

The most reliable way to activate this endogenous opioid-mediated modulating system is by suggestion of pain relief or by intense emotion directed away from the pain-causing injury (e.g., during severe threat or an athletic competition). In fact, pain-relieving endogenous opioids are released following surgical procedures and in patients given a placebo for pain relief.

Pain-modulating circuits can enhance as well as suppress pain. Both pain-inhibiting and pain-facilitating neurons in the medulla project to and control spinal pain-transmission neurons. Because pain-transmission neurons can be activated by modulatory neurons, it is theoretically possible to generate a pain signal with no peripheral noxious stimulus. In fact, human functional imaging studies have demonstrated increased activity in this circuit during migraine headaches. A central circuit that facilitates pain could account for the finding that pain can be induced by suggestion or enhanced by expectation and provides a framework for understanding how psychological factors can contribute to chronic pain.



## NEUROPATHIC PAIN

Lesions of the peripheral or central nociceptive pathways typically result in a loss or impairment of pain sensation. Paradoxically, damage to or dysfunction of these pathways can also produce pain. For example, damage to peripheral nerves, as occurs in diabetic neuropathy, or to primary afferents, as in herpes zoster infection, can result in pain that is referred to the body region innervated by the damaged nerves. Pain may also be produced by damage to the central nervous system (CNS), for example, in some patients following trauma or vascular injury to the spinal cord, brainstem, or thalamic areas that contain central nociceptive pathways. Such pains are termed *neuropathic* and are often severe and resistant to standard treatments for pain.

Neuropathic pain typically has an unusual burning, tingling, or electric shock-like quality and may occur spontaneously, without any stimulus, or be triggered by very light touch. These features are rare in other types of pain. On examination, a sensory deficit is characteristically co-extensive with the area of the patient's pain. *Hyperpathia*, a greatly exaggerated pain response to innocuous or mild nociceptive stimuli, especially when applied repeatedly, is also characteristic of neuropathic pain; patients often complain that the very lightest moving stimulus evokes exquisite pain (allodynia). In this regard, it is of clinical interest that a topical preparation of 5% lidocaine in patch form is effective for patients with postherpetic neuralgia who have prominent allodynia.

A variety of mechanisms contribute to neuropathic pain. As with sensitized primary afferent nociceptors, damaged primary afferents, including nociceptors, become highly sensitive to mechanical stimulation and may generate impulses in the absence of stimulation. Increased sensitivity and spontaneous activity are due, in part, to an increased density of sodium channels in the damaged nerve fiber. Damaged primary afferents may also develop sensitivity to norepinephrine. Interestingly, spinal cord pain-transmission neurons cut off from their normal input may also become spontaneously active. Thus, both central and peripheral nervous system hyperactivity contribute to neuropathic pain.

**Sympathetically Maintained Pain** Patients with peripheral nerve injury occasionally develop spontaneous pain in or beyond the region innervated by the nerve. This pain is often described as having a burning quality. The pain typically begins after a delay of hours to days or even weeks and is accompanied by swelling of the extremity, periarticular bone loss, and arthritic changes in the distal joints. Early in the course of the condition, the pain may be relieved by a local anesthetic block of the sympathetic innervation to the affected extremity. Damaged primary afferent nociceptors acquire adrenergic sensitivity and can be activated by stimulation of the sympathetic outflow. This constellation of spontaneous pain and signs of sympathetic dysfunction following injury has been termed *complex regional pain syndrome* (CRPS) **(Chap. 19)**. When this occurs after an identifiable nerve injury, it is termed CRPS type II (also known as posttraumatic neuralgia or, if severe, *causalgia*). When a similar clinical picture appears without obvious nerve injury, it is termed CRPS type I (also known as *reflex sympathetic dystrophy*). CRPS can be produced by a variety of injuries, including fractures of bone, soft tissue trauma, myocardial infarction, and stroke. CRPS type I typically resolves with symptomatic treatment; however, when it persists, detailed examination often reveals evidence of peripheral nerve injury. Although the pathophysiology of CRPS is poorly understood, the pain and the signs of inflammation, when acute, can be rapidly relieved by blocking the sympathetic nervous system. This implies that sympathetic activity can activate undamaged nociceptors when inflammation is present. Signs of sympathetic hyperactivity should be sought in patients with posttraumatic pain and inflammation and no other obvious explanation.

# TREATMENT

## Acute Pain

The ideal treatment for any pain is to remove the cause; thus, while treatment can be initiated immediately, efforts to establish the underlying etiology should always proceed as treatment begins. Sometimes, treating the underlying condition does not immediately relieve pain. Furthermore, some conditions are so painful that rapid and effective analgesia is essential (e.g., the postoperative state, burns, trauma, cancer, or sickle cell crisis). Analgesic medications are a first line of treatment in these cases, and all practitioners should be familiar with their use.

### ASPIRIN, ACETAMINOPHEN, AND NONSTEROIDAL ANTI-INFLAMMATORY AGENTS (NSAIDS)

These drugs are considered together because they are used for similar problems and may have a similar mechanism of action **(Table 14-1)**. All these compounds inhibit cyclooxygenase (COX), and except for acetaminophen, all have anti-inflammatory actions, especially at higher dosages. They are particularly effective for mild to moderate headache and for pain of musculoskeletal origin.

Because they are effective for these common types of pain and are available without prescription, COX inhibitors are by far the most used analgesics. They are absorbed well from the gastrointestinal tract and, with occasional use, have only minimal side effects. With chronic use, gastric irritation is a common side effect of aspirin and NSAIDs and is the problem that most frequently limits the dose that can be given. Gastric irritation is most severe with aspirin, which may cause erosion and ulceration of the gastric mucosa leading to bleeding or perforation. Because aspirin irreversibly acetylates platelet COX and thereby interferes with coagulation of the blood, gastrointestinal bleeding is a particular risk. Older age and history of gastrointestinal disease increase the risks of aspirin and NSAIDs. In addition to the well-known gastrointestinal toxicity of NSAIDs, nephrotoxicity is a significant problem for patients using these drugs on a chronic basis. Patients at risk for renal insufficiency, particularly those with significant contraction of their intravascular volume as occurs with chronic diuretic use or acute hypovolemia, should avoid NSAIDs. NSAIDs can also increase blood pressure in some individuals. Long-term treatment with NSAIDs requires regular blood pressure monitoring and treatment if necessary. Although toxic to the liver when taken in high doses, acetaminophen rarely produces gastric irritation and does not interfere with platelet function.

The introduction of parenteral forms of NSAIDs, ketorolac and diclofenac, extends the usefulness of this class of compounds in the management of acute severe pain. Both agents are sufficiently potent and rapid in onset to supplant opioids as first-line treatment for many patients with acute severe headache and musculoskeletal pain.

There are two major classes of COX: COX-1 is constitutively expressed, and COX-2 is induced in the inflammatory state.

**TABLE 14-1 Drugs for Relief of Pain**

| GENERIC NAME | DOSE, mg | INTERVAL | COMMENTS |
|---|---|---|---|
| **Nonnarcotic Analgesics: Usual Doses and Intervals** | | | |
| Acetylsalicylic acid | 650 PO | q4h | Enteric-coated preparations available |
| Acetaminophen | 650 PO | q4h | Side effects uncommon |
| Ibuprofen | 400 PO | q4–6h | Available without prescription |
| Naproxen | 250–500 PO | q12h | Naproxen is the common NSAID that poses the least cardiovascular risk, but it has a somewhat higher incidence of gastrointestinal bleeding |
| Fenoprofen | 200 PO | q4–6h | Contraindicated in renal disease |
| Indomethacin | 25–50 PO | q8h | Gastrointestinal side effects common |
| Ketorolac | 15–60 IM/IV | q4–6h | Available for parenteral use |
| Celecoxib | 100–200 PO | q12–24h | Useful for arthritis |
| Valdecoxib | 10–20 PO | q12–24h | Removed from U.S. market in 2005 |

| GENERIC NAME | PARENTERAL DOSE, mg | PO DOSE, mg | COMMENTS |
|---|---|---|---|
| **Narcotic Analgesics: Usual Doses and Intervals** | | | |
| Codeine | 30–60 q4h | 30–60 q4h | Nausea common |
| Oxycodone | — | 5–10 q4–6h | Usually available with acetaminophen or aspirin |
| Oxycodone extended-release | — | 10-40 q12h | Oral extended-release tablet; high potential for misuse |
| Morphine | 5 q4h | 30 q4h | |
| Morphine sustained release | — | 15–60 bid to tid | Oral slow-release preparation |
| Hydromorphone | 1–2 q4h | 2–4 q4h | Shorter acting than morphine sulfate |
| Levorphanol | 2 q6–8h | 4 q6–8h | Longer acting than morphine sulfate; absorbed well PO |
| Methadone | 5–10 q6–8h | 5–20 q6–8h | Due to long half-life, respiratory depression and sedation may persist after analgesic effect subsides; therapy should not be initiated with >40 mg/d, and dose escalation should be made no more frequently than every 3 days |
| Meperidine | 50–100 q3–4h | 300 q4h | Poorly absorbed PO; normeperidine is a toxic metabolite; routine use of this agent is not recommended |
| Butorphanol | — | 1–2 q4h | Intranasal spray |
| Fentanyl | 25–100 μg/h | — | 72-h transdermal patch |
| Buprenorphine | 5–20 μg/h | | 7-day transdermal patch |
| Buprenorphine | 0.3 q6–8h | | Parenteral administration |
| Tramadol | — | 50–100 q4–6h | Mixed opioid/adrenergic action |

| GENERIC NAME | UPTAKE BLOCKADE | | SEDATIVE POTENCY | ANTICHOLINERGIC POTENCY | ORTHOSTATIC HYPOTENSION | CARDIAC ARRHYTHMIA | AVERAGE DOSE, mg/d | RANGE, mg/d |
|---|---|---|---|---|---|---|---|---|
| | 5-HT | NE | | | | | | |
| **Antidepressants[a]** | | | | | | | | |
| Doxepin | ++ | + | High | Moderate | Moderate | Less | 200 | 75–400 |
| Amitriptyline | ++++ | ++ | High | Highest | Moderate | Yes | 150 | 25–300 |
| Imipramine | ++++ | ++ | Moderate | Moderate | High | Yes | 200 | 75–400 |
| Nortriptyline | +++ | ++ | Moderate | Moderate | Low | Yes | 100 | 40–150 |
| Desipramine | +++ | ++++ | Low | Low | Low | Yes | 150 | 50–300 |
| Venlafaxine | +++ | ++ | Low | None | None | No | 150 | 75–400 |
| Duloxetine | +++ | +++ | Low | None | None | No | 40 | 30–60 |

| GENERIC NAME | PO DOSE, mg | INTERVAL | COMMENTS |
|---|---|---|---|
| **Anticonvulsants and Antiarrythmics[a]** | | | |
| Carbamazepine | 200–300 | q6h | Rare aplastic anemia, GI irritation, hepatoitoxicity |
| Oxcarbamazepine | 300 | bid | Similar to carbamazepine |
| Gabapentin[b] | 600–1200 | q8h | Dizziness, GI irritation; useful in trigeminal neuralgia |
| Pregabalin | 150–600 | bid | Similar to gabapentin; dry mouth, edema |

[a]Antidepressants, anticonvulsants, and antiarrhythmics have not been approved by the U.S. Food and Drug Administration (FDA) for the treatment of pain. [b]Gabapentin in doses up to 1800 mg/d is FDA approved for postherpetic neuralgia.

*Abbreviations:* GI, gastrointestinal; 5-HT, serotonin; NE, norepinephrine; NSAID, nonsteroidal anti-inflammatory agent.

COX-2-selective drugs have similar analgesic potency and produce less gastric irritation than the nonselective COX inhibitors. The use of COX-2-selective drugs does not appear to lower the risk of nephrotoxicity compared to nonselective NSAIDs. On the other hand, COX-2-selective drugs offer a significant benefit in the management of acute postoperative pain because they do not affect blood coagulation. Nonselective COX inhibitors (especially aspirin) are usually contraindicated postoperatively because they impair platelet-mediated blood clotting and are thus associated with increased bleeding at the operative site. COX-2 inhibitors, including celecoxib (Celebrex), are associated with increased cardiovascular risk, including cardiovascular death, myocardial infarction, stroke, heart failure, or a thromboembolic event. It appears that this is a class effect of NSAIDs, excluding aspirin. These drugs are

contraindicated in patients in the immediate period after coronary artery bypass surgery and should be used with caution in elderly patients and those with a history of or significant risk factors for cardiovascular disease.

## OPIOID ANALGESICS

Opioids are the most potent pain-relieving drugs currently available. Of all analgesics, they have the broadest range of efficacy and provide the most reliable and effective treatment for rapid pain relief. Although side effects are common, most are reversible: nausea, vomiting, pruritus, sedation, and constipation are the most frequent and bothersome side effects. Respiratory depression is uncommon at standard analgesic doses but can be life-threatening. Opioid-related side effects can be reversed rapidly with the narcotic antagonist naloxone. Many physicians, nurses, and patients have a certain trepidation about using opioids that is based on a fear of initiating addiction in their patients. There is a small chance of patients becoming addicted to narcotics as a result of their appropriate medical use. For chronic pain, particularly chronic noncancer pain, the risk of addiction in patients taking opioids on a chronic basis remains small, and the risk appears to increase with dose escalation and duration of treatment. The physician should not hesitate to use opioid analgesics in patients with acute severe pain. Table 14-1 lists the most commonly used opioid analgesics.

Opioids produce analgesia by actions in the CNS. They activate pain-inhibitory neurons and directly inhibit pain-transmission neurons. Most of the commercially available opioid analgesics act at the same opioid receptor (μ-receptor), differing mainly in potency, speed of onset, duration of action, and optimal route of administration. Some side effects are due to accumulation of nonopioid metabolites that are unique to individual drugs. One striking example of this is normeperidine, a metabolite of meperidine. At higher doses of meperidine, typically >1 g/d, accumulation of normeperidine can produce hyperexcitability and seizures that are not reversible with naloxone. Normeperidine accumulation is increased in patients with renal failure.

The most rapid pain relief is obtained by intravenous administration of opioids; relief with oral administration is significantly slower. Because of the potential for respiratory depression, patients with any form of respiratory compromise must be kept under close observation following opioid administration; an oxygen-saturation monitor may be useful, but only in a setting where the monitor is under constant surveillance. Opioid-induced respiratory depression is primarily manifest as a reduction in respiratory rate and is typically accompanied by sedation. A fall in oxygen saturation represents a critical level of respiratory depression and the need for immediate intervention to prevent life-threatening hypoxemia. Newer monitoring devices that incorporate capnography or pharyngeal air flow can detect apnea at the point of onset and are now commonly used in hospitalized patients. Ventilatory assistance should be maintained until the opioid-induced respiratory depression has resolved. The opioid antagonist naloxone should be readily available whenever opioids are used at high doses or in patients with compromised pulmonary function. Opioid effects are dose-related, and there is great variability among patients in the doses that relieve pain and produce side effects. Synergistic respiratory depression is common when opioids are administered with other CNS depressants. Co-administration of benzodiazepines is particularly likely to produce respiratory depression and should be avoided, especially in outpatient pain management. Because of this variability in patient response, initiation of therapy requires titration to optimal dose and interval. The most important principle is to provide adequate pain relief. This requires determining whether the drug has adequately relieved the pain and timely reassessment to determine the optimal interval for dosing. *The most common error made by physicians in managing severe pain with opioids is to prescribe an inadequate dose. Because many patients are reluctant to complain, this practice leads to needless suffering.* In the absence of sedation at the expected time of peak effect, a physician should not hesitate to repeat the initial dose to achieve satisfactory pain relief.

A now standard approach to the problem of achieving adequate pain relief is the use of patient-controlled analgesia (PCA). PCA uses a microprocessor-controlled infusion device that can deliver a baseline continuous dose of an opioid drug as well as preprogrammed additional doses whenever the patient pushes a button. The patient can then titrate the dose to the optimal level. This approach is used most extensively for the management of postoperative pain, but there is no reason it should not be used for any hospitalized patient with persistent severe pain. PCA is also used for short-term home care of patients with intractable pain, such as that caused by metastatic cancer.

It is important to understand that the PCA device delivers small, repeated doses to maintain pain relief; in patients with severe pain, the pain must first be brought under control with a loading dose before transitioning to the PCA device. The bolus dose of the drug (typically 1 mg of morphine, 0.2 mg of hydromorphone, or 10 μg of fentanyl) can then be delivered repeatedly as needed. To prevent overdosing, PCA devices are programmed with a lockout period after each demand dose is delivered (typically starting at 10 min) and a limit on the total dose delivered per hour. Although some have advocated the use of a simultaneous continuous or basal infusion of the PCA drug, this may increase the risk of respiratory depression and has not been shown to increase the overall efficacy of the technique.

The availability of new routes of administration has extended the usefulness of opioid analgesics. Most important is the availability of spinal administration. Opioids can be infused through a spinal catheter placed either intrathecally or epidurally. By applying opioids directly to the spinal or epidural space adjacent to the spinal cord, regional analgesia can be obtained using relatively low total doses. Indeed, the dose required to produce effective analgesia when using morphine intrathecally (0.1–0.3 mg) is a fraction of that required to produce similar analgesia when administered intravenously (5–10 mg). In this way, side effects such as sedation, nausea, and respiratory depression can be minimized. This approach has been used extensively during labor and delivery and for postoperative pain relief following surgical procedures. Continuous intrathecal delivery via implanted spinal drug-delivery systems is now commonly used, particularly for the treatment of cancer-related pain that would require sedating doses for adequate pain control if given systemically. Opioids can also be given intranasally (butorphanol), rectally, and transdermally (fentanyl and buprenorphine), or through the oral mucosa (fentanyl), thus avoiding the discomfort of frequent injections in patients who cannot be given oral medication. The fentanyl and buprenorphine transdermal patches have the advantage of providing steady plasma levels, which may improve patient comfort.

Recent additions to the armamentarium for treating opioid-induced side effects are the peripherally acting opioid antagonists alvimopan (Entereg) and methylnaltrexone (Rellistor). Alvimopan is available as an orally administered agent that is restricted to the intestinal lumen by limited absorption; methylnaltrexone is available in a subcutaneously administered form that has virtually no penetration into the CNS. Both agents act by binding to peripheral μ-receptors, thereby inhibiting or reversing the effects of opioids at these peripheral sites. The action of both agents is restricted to receptor sites outside of the CNS; thus, these drugs can reverse the adverse effects of opioid analgesics that are mediated through their peripheral receptors without reversing their CNS-mediated analgesic effects. Alvimopan has proven effective in lowering the duration of persistent ileus following abdominal surgery in patients receiving opioid analgesics for postoperative pain control. Methylnaltrexone has proven effective for relief of opioid-induced constipation in patients taking opioid analgesics on a chronic basis.

***Opioid and COX Inhibitor Combinations*** When used in combination, opioids and COX inhibitors have additive or synergistic

effects. Because a lower dose of each can be used to achieve the same degree of pain relief and their side effects are typically dose related, such combinations are used to lower the severity of dose-related side effects. However, fixed-ratio combinations of an opioid with acetaminophen carry an important risk. Dose escalation because of increased severity of pain or decreased opioid effect as a result of tolerance may lead to ingestion of levels of acetaminophen that are toxic to the liver. Although acetaminophen-related hepatotoxicity is uncommon, it remains a significant cause for liver failure. Thus, many practitioners have moved away from the use of opioid-acetaminophen combination analgesics to avoid the risk of excessive acetaminophen exposure as the dose of the analgesic is escalated.

## CHRONIC PAIN

Managing patients with chronic pain is intellectually and emotionally challenging. Sensitization of the nervous system can occur without an obvious precipitating cause, e.g., fibromyalgia, or chronic headache. In many patients, chronic pain becomes a distinct disease unto itself. The pain-generating mechanism is often difficult or impossible to determine with certainty; such patients are demanding of the physician's time and often appear emotionally distraught. The traditional medical approach of seeking an obscure organic pathology is often unhelpful. On the other hand, psychological evaluation and behaviorally based treatment paradigms are frequently helpful, particularly in the setting of a multidisciplinary pain-management center. Unfortunately, this approach, while effective, remains largely underused in current medical practice.

There are several factors that can cause, perpetuate, or exacerbate chronic pain. First, of course, the patient may simply have a disease that is characteristically painful for which there is presently no cure. Arthritis, cancer, chronic daily headaches, fibromyalgia, and diabetic neuropathy are examples of this. Second, there may be secondary perpetuating factors that are initiated by disease and persist after that disease has resolved. Examples include damaged sensory nerves, sympathetic efferent activity, and painful reflex muscle contraction (spasm). Finally, a variety of psychological conditions can exacerbate or even cause pain.

There are certain areas to which special attention should be paid in a patient's medical history. Because depression **(Chap. 463)** is the most common emotional disturbance in patients with chronic pain, patients should be questioned about their mood, appetite, sleep patterns, and daily activity. A simple standardized questionnaire, such as the Beck Depression Inventory, can be a useful screening device. It is important to remember that major depression is a common, treatable, and potentially fatal illness.

Other clues that a significant emotional disturbance is contributing to a patient's chronic pain complaint include pain that occurs in multiple, unrelated sites; a pattern of recurrent, but separate, pain problems beginning in childhood or adolescence; pain beginning at a time of emotional trauma, such as the loss of a parent or spouse; a history of physical or sexual abuse; and past or present substance abuse.

On examination, special attention should be paid to whether the patient guards the painful area and whether certain movements or postures are avoided because of pain. Discovering a mechanical component to the pain can be useful both diagnostically and therapeutically. Painful areas should be examined for deep tenderness, noting whether this is localized to muscle, ligamentous structures, or joints. Chronic myofascial pain is very common, and in these patients, deep palpation may reveal highly localized trigger points that are firm bands or knots in muscle. Relief of the pain following injection of local anesthetic into these trigger points supports the diagnosis. A neuropathic component to the pain is indicated by evidence of nerve damage, such as sensory impairment, exquisitely sensitive skin (allodynia), weakness, and muscle atrophy, or loss of deep tendon reflexes. Evidence suggesting sympathetic nervous system involvement includes the presence of diffuse swelling, changes in skin color and temperature, and hypersensitive skin and joint tenderness compared with the normal side. Relief of the pain with a sympathetic block supports the diagnosis, but once the condition becomes chronic, the response to sympathetic blockade is of variable magnitude and duration; the role for repeated sympathetic blocks in the overall management of CRPS is unclear.

A guiding principle in evaluating patients with chronic pain is to assess both emotional and somatic causal and perpetuating factors before initiating therapy. Addressing these issues together, rather than waiting to address emotional issues after somatic causes of pain have been ruled out, improves compliance in part because it assures patients that a psychological evaluation does not mean that the physician is questioning the validity of their complaint. Even when a somatic cause for a patient's pain can be found, it is still wise to look for other factors. For example, a cancer patient with painful bony metastases may have additional pain due to nerve damage and may also be depressed. Optimal therapy requires that each of these factors be assessed and treated.

## TREATMENT

### Chronic Pain

Once the evaluation process has been completed and the likely causative and exacerbating factors identified, an explicit treatment plan should be developed. An important part of this process is to identify specific and realistic functional goals for therapy, such as getting a good night's sleep, being able to go shopping, or returning to work. A multidisciplinary approach that uses medications, counseling, physical therapy, nerve blocks, and even surgery may be required to improve the patient's quality of life. There are also some newer, minimally invasive procedures that can be helpful for some patients with intractable pain. These include image-guided interventions such as epidural injection of glucocorticoids for acute radicular pain and radiofrequency treatment of the facet joints for chronic facet-related back and neck pain. For patients with severe and persistent pain that is unresponsive to more conservative treatment, placement of electrodes on peripheral nerves or within the spinal canal on nerve roots or in the space overlying the dorsal columns of the spinal cord (spinal cord stimulation) has shown significant benefit. The criteria for predicting which patients will respond to these procedures continue to evolve. They are generally reserved for patients who have not responded to conventional pharmacologic approaches. Referral to a multidisciplinary pain clinic for a full evaluation should precede any invasive procedure. Such referrals are clearly not necessary for all chronic pain patients. For some, pharmacologic management alone can provide adequate relief.

#### ANTIDEPRESSANT MEDICATIONS

The tricyclic antidepressants (TCAs), particularly nortriptyline and desipramine (Table 14-1), are useful for the management of chronic pain. Although developed for the treatment of depression, the TCAs have a spectrum of dose-related biologic activities that include analgesia in a variety of chronic clinical conditions. Although the mechanism is unknown, the analgesic effect of TCAs has a more rapid onset and occurs at a lower dose than is typically required for the treatment of depression. Furthermore, patients with chronic pain who are not depressed obtain pain relief with antidepressants. There is evidence that TCAs potentiate opioid analgesia, so they may be useful adjuncts for the treatment of severe persistent pain such as occurs with malignant tumors. **Table 14-2** lists some of the painful conditions that respond to TCAs. TCAs are of particular value in the management of neuropathic pain such as occurs in diabetic neuropathy and postherpetic neuralgia, for which there are few other therapeutic options.

The TCAs that have been shown to relieve pain have significant side effects (Table 14-1; **Chap. 463**). Some of these side effects, such as orthostatic hypotension, drowsiness, cardiac conduction delay, memory impairment, constipation, and urinary retention,

**TABLE 14-2 Painful Conditions That Respond to Tricyclic Antidepressants**

| |
|---|
| Postherpetic neuralgia[a] |
| Diabetic neuropathy[a] |
| Fibromyalgia[a] |
| Tension headache[a] |
| Migraine headache[a] |
| Rheumatoid arthritis[a,b] |
| Chronic low back pain[b] |
| Cancer |
| Central poststroke pain |

[a]Controlled trials demonstrate analgesia. [b]Controlled studies indicate benefit but not analgesia.

are particularly problematic in elderly patients, and several are additive to the side effects of opioid analgesics. The selective serotonin reuptake inhibitors such as fluoxetine (Prozac) have fewer and less serious side effects than TCAs, but they are much less effective for relieving pain. In contrast, venlafaxine (Effexor) and duloxetine (Cymbalta), which are nontricyclic antidepressants that block both serotonin and norepinephrine reuptake, appear to retain most of the pain-relieving effects of TCAs with a side effect profile more like that of the selective serotonin reuptake inhibitors. Because of their favorable adverse effect profiles, these drugs have largely supplanted the use of TCAs in treatment of chronic pain.

## ANTICONVULSANTS AND ANTIARRHYTHMICS

These drugs are useful primarily for patients with neuropathic pain. Phenytoin (Dilantin) and carbamazepine (Tegretol) were first shown to relieve the pain of trigeminal neuralgia **(Chap. 452)**. This pain has a characteristic brief, shooting, electric shock-like quality. In fact, anticonvulsants seem to be particularly helpful for pains that have such a lancinating quality. Newer anticonvulsants, the calcium channel alpha-2-delta subunit ligands gabapentin (Neurontin) and pregabalin (Lyrica), are effective for a broad range of neuropathic pains. Furthermore, because of their favorable side effect profiles, these newer anticonvulsants are often used as first-line agents.

## CANNABINOIDS

These agents are widely used for their analgesic properties, although published evidence suggests that any analgesic effects are modest, with small increases in pain threshold reported and variable reductions in clinical pain intensity. Cannabis more consistently reduces the unpleasantness of the pain experience and, in cancer-related pain, can lessen the nausea and vomiting associated with chemotherapy use. **Marijuana and related compounds are discussed in Chap. 466.**

## CHRONIC OPIOID MEDICATION

The long-term use of opioids is accepted for patients with pain due to malignant disease. Although opioid use for chronic pain of nonmalignant origin is controversial, for many patients, opioids are the only option that produces meaningful pain relief. This is understandable because opioids are the most potent and have the broadest range of efficacy of any analgesic medications. Although addiction is rare in patients who first use opioids for pain relief, some degree of tolerance and physical dependence is likely with long-term use. Furthermore, studies suggest that long-term opioid therapy may worsen pain in some individuals, termed *opioid-induced hyperalgesia*. Therefore, before embarking on opioid therapy, other options should be explored, and the limitations and risks of opioids should be explained to the patient. It is also important to recognize that some opioid analgesic medications have mixed agonist-antagonist properties (e.g., butorphanol and buprenorphine). From a practical standpoint, this means that they may worsen pain by inducing an abstinence syndrome in patients who are actively being treated with other opioids and are physically dependent.

With long-term outpatient use of orally administered opioids, it may be desirable to use long-acting compounds such as levorphanol, methadone, extended-release morphine or oxycodone, or transdermal fentanyl (Table 14-1). The pharmacokinetic profiles of these drug preparations enable the maintenance of sustained analgesic blood levels, potentially minimizing side effects such as sedation that are associated with high peak plasma levels and reducing the likelihood of rebound pain associated with a rapid fall in plasma opioid concentration. Extended-release opioid formulations are approved primarily for patients who are already taking other opioids and should not be used as first-line opioids for pain. Although long-acting opioid preparations may provide superior pain relief in patients with a continuous pattern of ongoing pain, others suffer from intermittent severe episodic pain and experience superior pain control and fewer side effects with the periodic use of short-acting opioid analgesics. Constipation is a virtually universal side effect of opioid use and should be treated expectantly. As noted earlier in the discussion of acute pain treatment, an advance for patients is the development of peripherally acting opioid antagonists that can reverse the constipation associated with opioid use without interfering with analgesia.

Soon after the introduction of an extended-release oxycodone formulation (OxyContin) in the late 1990s, a dramatic rise in emergency department visits and deaths associated with oxycodone ingestion appeared. This appears to be due primarily to individuals using a prescription opioid nonmedically. Drug-induced deaths have rapidly risen and are now the second leading cause of death in Americans, just behind motor vehicle fatalities. In 2011, the Office of National Drug Control Policy established a multifaceted approach to address prescription drug abuse, including prescription drug monitoring programs (PDMPs) that allow practitioners to determine if patients are receiving prescriptions from multiple providers and use of law enforcement to eliminate improper prescribing practices. In 2016, the Centers for Disease Control and Prevention (CDC) released the *CDC Guideline for Prescribing Opioids for Chronic Pain*, with recommendations for primary care clinicians who are prescribing opioids for chronic noncancer pain. A modified approach to opioid prescribing was published in 2019 by the Health and Human Services Task Force on chronic pain best medical practices and updated in 2022. These guidelines address (1) when to initiate or continue opioids for chronic pain; (2) opioid selection, dosage, duration, follow-up, and discontinuation; and (3) assessing risk and addressing harms of opioid use. The recent increase in scrutiny leaves many practitioners hesitant to prescribe opioid analgesics, other than for brief periods to control pain associated with illness or injury. For now, the choice to begin chronic opioid therapy for a given patient is left to the individual practitioner. The new CDC guidelines aim to help practitioners reduce the risk of opioid misuse and overdose deaths while providing effective pain management for their patients. Pragmatic guidelines for properly selecting and monitoring patients receiving chronic opioid therapy are shown in **Table 14-3**; a checklist for primary care clinicians prescribing opioids for noncancer pain is shown in **Table 14-4**. Practitioners should refer to the 2022 Clinical Practice Guideline for prescribing opioids for pain (referenced in Further Reading, below). **Opioid use disorders are comprehensively discussed in Chap. 467.**

## TREATMENT OF NEUROPATHIC PAIN

It is important to individualize treatment for patients with neuropathic pain. Several general principles should guide therapy: the first is to move quickly to provide relief, and the second is to minimize drug side effects. For example, in patients with postherpetic neuralgia and significant cutaneous hypersensitivity, topical lidocaine (e.g. Lidoderm patches) can provide immediate relief without side effects. The anticonvulsants gabapentin or pregabalin

**TABLE 14-3 Guidelines for Selecting and Monitoring Patients Receiving Chronic Opioid Therapy (COT) for the Treatment of Chronic, Noncancer Pain**

**Patient Selection**

- Conduct a history, physical examination, and appropriate testing, including an assessment of risk of substance abuse, misuse, or addiction.
- Consider a trial of COT if pain is moderate or severe, pain is having an adverse impact on function or quality of life, and potential therapeutic benefits outweigh potential harms.
- A benefit-to-harm evaluation, including a history, physical examination, and appropriate diagnostic testing, should be performed and documented before and on an ongoing basis during COT.

**Informed Consent and Use of Management Plans**

- Informed consent should be obtained. A continuing discussion with the patient regarding COT should include goals, expectations, potential risks, and alternatives to COT.
- Consider using a written COT management plan to document patient and clinician responsibilities and expectations and assist in patient education.

**Initiation and Titration**

- Initial treatment with opioids should be considered as a therapeutic trial to determine whether COT is appropriate.
- Opioid selection, initial dosing, and titration should be individualized according to the patient's health status, previous exposure to opioids, attainment of therapeutic goals, and predicted or observed harms.

**Monitoring**

- Reassess patients on COT periodically and as warranted by changing circumstances. Monitoring should include documentation of pain intensity and level of functioning, assessments of progress toward achieving therapeutic goals, presence of adverse events, and adherence to prescribed therapies.
- In patients on COT who are at high risk or who have engaged in aberrant drug-related behaviors, clinicians should periodically obtain urine drug screens or other information to confirm adherence to the COT plan of care.
- In patients on COT not at high risk and not known to have engaged in aberrant drug-related behaviors, clinicians should consider periodically obtaining urine drug screens or other information to confirm adherence to the COT plan of care.

*Source:* Reproduced with permission from R Chou et al: Clinical guidelines for the use of chronic opioid therapy in chronic noncancer pain. J Pain 10:113-130, 2009.

**TABLE 14-4 Centers for Disease Control and Prevention Checklist for Prescribing Opioids for Chronic Pain**

**For Primary Care Providers Treating Adults (18+) with Chronic Pain ≥3 Months, Excluding Cancer, Palliative, and End-of-Life Care**

CHECKLIST

WHEN CONSIDERING LONG-TERM OPIOID THERAPY

- Set realistic goals for pain and function based on diagnosis (e.g., walk around the block).
- Check that nonopioid therapies tried and optimized.
- Discuss benefits and risks (e.g., addiction, overdose) with patient.
- Evaluate risk of harm or misuse.
  - Discuss risk factors with patient.
  - Check prescription drug monitoring program (PDMP) data.
  - Check urine drug screen.
- Set criteria for stopping or continuing opioids.
- Assess baseline pain and function (e.g., Pain, Enjoyment, General Activity [PEG] scale).
- Schedule initial reassessment within 1–4 weeks.
- Prescribe short-acting opioids using lowest dosage on product labeling; match duration to scheduled reassessment.

IF RENEWING WITHOUT A PATIENT VISIT

- Check that return visit is scheduled ≤3 months from last visit.

WHEN REASSESSING AT A PATIENT VISIT

- Continue opioids only after confirming clinically meaningful improvements in pain and function without significant risks or harm.
- Assess pain and function (e.g., PEG); compare results to baseline.
- Evaluate risk of harm or misuse:
  - Observe patient for signs of oversedation or overdose risk. If yes: Taper dose.
  - Check PDMP.
  - Check for opioid use disorder if indicated (e.g., difficulty controlling use). If yes: Refer for treatment.
- Check that nonopioid therapies optimized. Determine whether to continue, adjust, taper, or stop opioids.
- Calculate opioid dosage morphine milligram equivalent (MME).
  - If ≥50 MME/day total (≥50 mg hydrocodone; ≥33 mg oxycodone), increase frequency of follow-up; consider offering naloxone.
  - Avoid ≥90 MME/day total (≥90 mg hydrocodone; ≥60 mg oxycodone), or carefully justify; consider specialist referral.
- Schedule reassessment at regular intervals (≤3 months).

*Source:* Centers for Disease Control and Prevention, available at *https://stacks.cdc.gov/view/cdc/38025*. Accessed February 18, 2024 (Public Domain).

(see above) or antidepressants (nortriptyline, desipramine, duloxetine, or venlafaxine) can be used as first-line drugs for patients with neuropathic pain. Systemically administered antiarrhythmic drugs such as lidocaine and mexiletine are less likely to be effective. Although intravenous infusion of lidocaine can provide analgesia for patients with different types of neuropathic pain, the relief is usually transient, typically lasting just hours after the cessation of the infusion. The oral lidocaine congener mexiletine is poorly tolerated, producing frequent gastrointestinal adverse effects. There is no consensus on which class of drug should be used as a first-line treatment for any chronically painful condition. However, because relatively high doses of anticonvulsants are required for pain relief, sedation is not uncommon. Sedation is also a problem with TCAs but is much less of a problem with serotonin/norepinephrine reuptake inhibitors (SNRIs; e.g., venlafaxine and duloxetine). Thus, in the elderly or in patients whose daily activities require high-level mental activity, these drugs should be considered the first line. In contrast, opioid medications should be used as a second- or third-line drug class. Although highly effective for many painful conditions, opioids are sedating, and their effect tends to lessen over time, leading to dose escalation and, occasionally, a worsening of pain. A couple of interesting alternatives to pure opioids are two drugs with mixed opioid and norepinephrine reuptake action: tramadol and tapentadol. Tramadol is a relatively weak opioid but is sometimes effective for pain unresponsive to nonopioid analgesics. Tapentadol is a stronger opioid, but its analgesic action is apparently enhanced by the norepinephrine reuptake blockade. Similarly, drugs of different classes can be used in combination to optimize pain control.

It is worth emphasizing that many patients, especially those with chronic pain, seek medical attention primarily because they are suffering and because only physicians can provide the medications required for pain relief. A primary responsibility of all physicians is to minimize the physical and emotional discomfort of their patients. Familiarity with pain mechanisms and analgesic medications is an important step toward accomplishing this aim.

## FURTHER READING


De Vita MJ et al: Association of cannabinoid administration with experimental pain in healthy adults a systematic review and meta-analysis. JAMA Psychiatry 75:1118, 2018.

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# 15 Chest Discomfort

David A. Morrow



Chest discomfort is among the most common reasons for which patients present for medical attention at either an emergency department (ED) or an outpatient clinic. The evaluation of nontraumatic chest discomfort is inherently challenging owing to the broad variety of possible causes, a minority of which are life-threatening conditions that should not be missed. It is helpful to frame the initial diagnostic assessment and triage of patients with acute chest discomfort around three categories: (1) myocardial ischemia; (2) other cardiopulmonary causes (myopericardial disease, aortic emergencies, and pulmonary conditions); and (3) noncardiopulmonary causes. Although rapid identification of high-risk conditions is a priority of the initial assessment, strategies that incorporate routine liberal use of testing carry the potential for adverse effects of unnecessary investigations.

## EPIDEMIOLOGY AND NATURAL HISTORY

Chest discomfort is one of the three most common reason for visits to the ED in the United States, resulting in 6–7 million emergency visits each year. More than 60% of patients with this presentation are hospitalized for further testing, and most of the remainder undergo additional investigation in the ED. Fewer than 15% of evaluated patients are eventually diagnosed with acute coronary syndrome (ACS), with rates of 10–20% in most series of unselected populations, and a rate as low as 5% in some studies. The most common diagnoses are gastrointestinal causes **(Fig. 15-1)**, and as few as 5% are other life-threatening cardiopulmonary conditions. In a large proportion of patients with transient acute chest discomfort, ACS or another acute cardiopulmonary cause is excluded but the cause is not determined. Therefore, the resources and time devoted to the evaluation of chest discomfort *in the absence of a severe cause* are substantial. Nevertheless, historically, a disconcerting 2–6% of patients with chest discomfort of presumed nonischemic etiology who are discharged from the ED were later deemed to have had a missed myocardial infarction (MI).

The natural histories of ACS, myocarditis, acute pericardial diseases, pulmonary embolism, and aortic emergencies are discussed in **Chaps. 281, 284, 285, 286, 290, and 291**, respectively. The estimated rate of major cardiovascular events through 30 days in patients with acute chest pain who had been stratified as low risk was 2.5% in a large population-based study that excluded patients with ST-segment elevation or definite noncardiac chest pain.

## CAUSES OF CHEST DISCOMFORT

The major etiologies of chest discomfort are discussed in this section and summarized in **Table 15-1**. Additional elements of the history, physical examination, and diagnostic testing that aid in distinguishing these causes are discussed in a later section (see "Approach to the Patient").

### MYOCARDIAL ISCHEMIA/INJURY

Myocardial ischemia causing chest discomfort, termed *angina pectoris*, is a primary clinical concern in patients presenting with chest symptoms. Myocardial ischemia is precipitated by an imbalance between myocardial oxygen requirements and myocardial oxygen supply, resulting in insufficient delivery of oxygen to meet the heart's metabolic demands. Myocardial oxygen consumption may be elevated by increases in heart rate, ventricular wall stress, and myocardial contractility, whereas myocardial oxygen supply is determined by coronary blood flow and coronary arterial oxygen content. When myocardial ischemia is sufficiently severe and prolonged in duration (as little as 20 min), irreversible cellular injury occurs, resulting in MI.

The pathobiology of ischemic heart disease is described in **Chap. 284**. *Stable angina* is characterized by ischemic episodes that are typically precipitated by a superimposed increase in oxygen demand during physical exertion and relieved upon resting. Unstable ischemic heart disease **(Chaps. 285 and 286)** is classified clinically by the presence or absence of detectable acute myocardial injury and the presence or absence of ST-segment elevation on the patient's electrocardiogram (ECG). Unstable ischemic heart disease is classified as *unstable angina* when there is no detectable acute myocardial injury and either as *non–ST elevation MI* (NSTEMI; see **Chap. 285**) or *ST-elevation MI* (STEMI; see **Chap. 286**) when there is evidence of acute myocardial injury.

Clinicians should be aware that unstable ischemic symptoms may also occur predominantly because of increased myocardial oxygen demand (e.g., during intense psychological stress) or because of decreased oxygen delivery due to anemia, hypoxia, or hypotension. However, the term *acute coronary syndrome*, which encompasses unstable angina, NSTEMI, and STEMI, is in general reserved for ischemia precipitated by acute coronary atherothrombosis. To guide therapeutic strategies, a standardized system for classification of MI has been expanded to discriminate MI resulting from acute coronary thrombosis (type 1 MI) from MI occurring secondary to other imbalances of myocardial oxygen supply and demand (type 2 MI; see **Chap. 285**). Conditions associated with extreme myocardial oxygen demand and impaired endocardial blood flow, such as aortic valve disease **(Chap. 291)**, hypertrophic cardiomyopathy, or idiopathic dilated cardiomyopathy **(Chaps. 266–270)**, can precipitate myocardial ischemia in patients with or without underlying obstructive atherosclerosis. Ischemic causes of chest pain and myocardial injury are additionally distinguished from nonischemic causes of acute myocardial injury, such as myocarditis.

**Characteristics of Ischemic Chest Discomfort** The clinical characteristics of angina pectoris, often referred to simply as "angina," are highly similar in quality and location whether the ischemic discomfort is a manifestation of stable ischemic heart disease, unstable angina, or MI (Table 15-1). Heberden initially described angina as a sense of

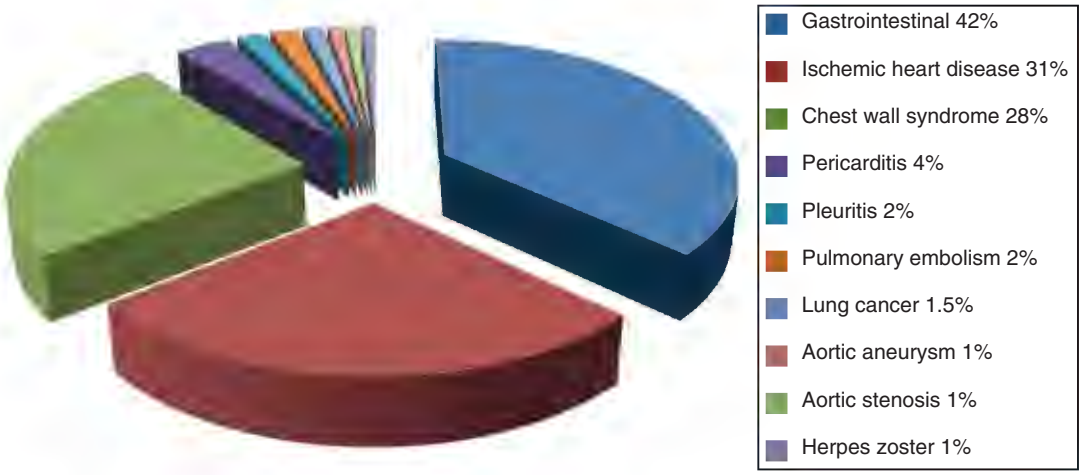


**FIGURE 15-1 Distribution of final discharge diagnoses in patients with nontraumatic acute chest pain.** *(Figure prepared from data in P Fruergaard et al: Eur Heart J 17:1028, 1996.)*